Zoo Animal Tracker - Kotlin Program

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1. Aim

To create a Zoo Animal Tracker using Kotlin that demonstrates inheritance, abstract classes, interfaces, and polymorphism.

2. Concepts Used

- Abstract Class
- Inheritance
- Interface
- Polymorphism
- Collections in Kotlin

3. Program Code

```
// Interface
interface SoundMaker {
    fun makeSound()
}

// Abstract Class
abstract class Animal(val name: String, val age: Int) {
    abstract fun performAction()

    fun displayInfo() {
        println("Name: $name | Age: $age")
    }
}

// Lion Class
class Lion(name: String, age: Int) : Animal(name, age), SoundMaker {

    override fun makeSound() {
        println("$name roars loudly! □")
    }
```

```
    override fun performAction() {
        println("$name is hunting in the zoo.")
    }
}

// Elephant Class
class Elephant(name: String, age: Int) : Animal(name, age), SoundMaker
{

    override fun makeSound() {
        println("$name trumpets loudly! □")
    }

    fun sprayWater() {
        println("$name sprays water with its trunk! □")
    }

    override fun performAction() {
        sprayWater()
    }
}

// Parrot Class
class Parrot(name: String, age: Int) : Animal(name, age), SoundMaker {

    override fun makeSound() {
        println("$name says: Hello! □")
    }

    fun mimicWords(words: String) {
        println("$name mimics: $words")
    }

    override fun performAction() {
        mimicWords("Welcome to the Zoo!")
    }
}

// Main Function
fun main() {
    val zooAnimals: List<Animal> = listOf(
        Lion("Simba", 5),
        Elephant("Dumbo", 10),
        Parrot("Polly", 2)
    )

    println("□ --- Zoo Animal Tracker --- □\n")

    for (animal in zooAnimals) {
```

```
        animal.displayInfo()
        animal.performAction()

        if (animal is SoundMaker) {
            animal.makeSound()
        }

        println("----------------------------")
    }
}
```

4. Sample Output

```
□ --- Zoo Animal Tracker --- □

Name: Simba | Age: 5
Simba is hunting in the zoo.
Simba roars loudly! □
----------------------------

Name: Dumbo | Age: 10
Dumbo sprays water with its trunk! □
Dumbo trumpets loudly! □
----------------------------

Name: Polly | Age: 2
Polly mimics: Welcome to the Zoo!
Polly says: Hello! □
----------------------------
```

5. Conclusion

This program successfully demonstrates object-oriented programming principles such as inheritance, abstraction, interfaces, and polymorphism using Kotlin. The Zoo Animal Tracker simulates real-world animal behaviors in a structured and reusable class design.